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Resumen/Summary The purpose of this Technical Report is to define the way to use the URCO program. This program calculates the allowable values for fastened joints made in composite material for Groups III, IV and V, tape and fabric, according to the official EADS CASA procedure. It also calculates the reserve factors of the joint for a set of loads input by the user. The issue B of this report includes criteria for the determination of the geometrical parameters defining the joint. PROPIEDAD DE EADS CASA. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Proyectos y Sistemas de EADS CASA. EADS CASA PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorisation by EADS CASA Engineering and System Direction.				
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1 Technical sheet of the program

Technical Note NT-T-ADP-06003

<u>PROGRAM</u>	<u>REFERENCE</u>	<u>DATE</u>
URCO		2006

OBJECTIVE

This program calculates the allowable values for fastened joints made in composite as well as the reserve factors for a set of loads.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

2200

EXECUTION MODE

Interactive

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VERSION

4.1

ADDITIONAL DOCUMENTS

Technical Note:	NT-T-ADP-06004. Theoretical Manual of the URCO Program.
Technical Note:	NT-T-ADP-06005. Programmer's Manual of the URCO Program.

2 Introduction

This report makes up the User's Manual for version 4 of the URCO Program, which has been developed by EADS CASA from the year 1999 to date.

This program calculates the allowable values for fastened joints made in composite material for Groups III, IV and V, tape and fabric, according to the official EADS CASA procedure.

These allowable values are the following:

- **A:** It means the tension or compression strength in the cross section at the hole location of the laminate for the by-pass load only.
- **B:** It is the tension strength in the cross section at the hole location for the applied load in the fastener.
- **C:** It is the bearing strength of the laminate submitted only to the fastener load.

It also calculates the reserve factors of the joint for a set of loads input by the user. These reserve factors are the following:

- Reserve factor for failure of the cross section at the hole location due to tension or compression.
- Reserve factor for bearing failure.

Differences between versions:

- Version 1.0:

Initial version developed in Fortran 77.

In parallel with this version, a Windows application was issued for interactive input and output. This application works interactively with the user. The analysis is performed by means of a call to the original Fortran program. The data interchange is done through temporary files, deleted at the end of the execution.

- Version 2.0:

The Graphic User Interface (GUI) is modified, reducing its size and including figures to clarify the input and output data meaning.

The user can choose the language: Spanish or English.

The program is restructured to allow the creation of a dynamic link library DLL for Windows operative system.

- Version 3.0:

The Fortran version is fully rearranged.

The interpolation method in the internal data is modified. In previous versions, the interpolation was done by “splines”, calling to routines included in the IMSL library. In this version, a linear interpolation is performed. This fact eliminates the dependency from the IMSL software.

The internal data is extrapolated, avoiding some functional limitations of the program and advising with appropriate warnings to the user when the considered data were extrapolated.

The Graphic User Interface includes now the possibility to directly send the problem results to the printer.

- Version 3.1:

The program version is explicitly included at the bottom of the output file.

- Version 4.0:

The internal data of the program corresponding to tape of Group V material are modified according to Technical Report NT-0-ADP-04001.

- Version 4.1:

The way to calculate the effective distance to the end has been modified. In previous version was calculated as:

- Minimum value of: Distance to the plate end, measured in the load direction.
Fasteners pitch, measured in the load direction.

$$e_{\text{eff}} = \text{Minimum}(e, p)$$

Now, it is calculated as:

- Minimum value of: Distance to the plate end, measured in the load direction.
Fasteners pitch less a hole radius, measured in the load direction.

$$e_{\text{eff}} = \text{Minimum}(e, p - D / 2)$$

3 Execution of the program

This program can be executed by typing its **full path name**. If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table), it can be executed by typing the command:

\$urco

The program asks for the language to use, the type of material and all required data to perform the analysis. The results will be presented on the screen. Additionally it is possible to specify an output file, as shown in the following example:

```
Idioma / Language:
=====
  1 ... ESPAÑOL
  2 ... ENGLISH
Idioma / Language: 2
.....
.....
.....
Do you want to print the results in an file? ..... (Y/N): Y
Name of the output results file: Example.txt
```

Details of the interactive data flow on the screen are provided in the example included in this report. The CPU time for calculation is negligible.

4 Input data

All input data for the program have to be provided in interactive mode. They are enumerated and explained below:

- **Language:**

The user can select Spanish or English in the following screen:

```
Idioma / Language:
=====

1 ... ESPAÑOL

2 ... ENGLISH

Idioma / Language: 2
```

- **Material and lay-up:**

```
MATERIAL:
=====

1 ... Fabric Group III
2 ... Tape   Group III
3 ... Fabric Group IV
4 ... Tape   Group IV
5 ... Fabric Group V
6 ... Tape   Group V

Type of material: 6

Percentage of layers at 0 degrees (%): 25

Percentage of layers at 45 degrees (%): 50
```

- **Type of joint:**

The user can select the type of joint for the element to be analysed, in the following screen:

```
TYPE OF JOINT:
=====

1 ... Single shear
2 ... Double shear

Type of joint: 1
```

The option 1 must be selected when the element to be analysed is in a single shear joint or it is an external plate in a double shear joint:

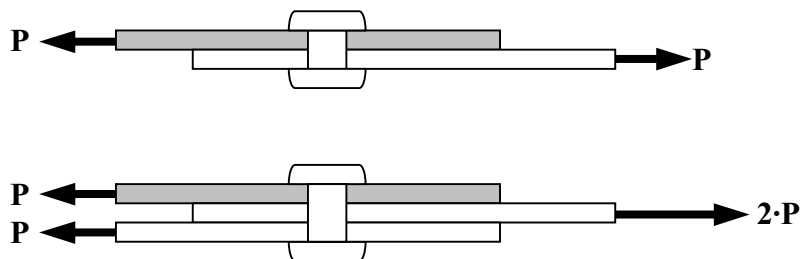


Figure 1: Elements in single shear.

The option 2 must be selected when the element to be analysed is the central one in a double shear joint:

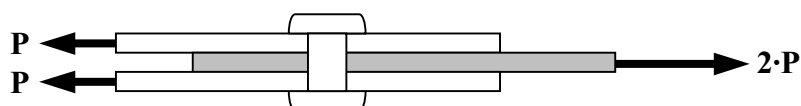


Figure 2: Element in double shear.

- **Type of fastener:**

TYPE OF FASTENER:

=====

- 1 ... Fastener of protuding head
- 2 ... Countersunk fastener

Type of fastener: 2

- **Geometry and loading:**

All dimensions must be expressed in “mm”. The fastener load must be in “N”.

Fastener diameter: 4

Effective width of the laminate: 16

Distance to the end of the joint in the load direction: 100

Laminate thickness: 4

Fastener pitch in the load direction: 16

By-pass load: + Tension
- Compression

Microstrains due to the by-pass load: 2000

Transferred load through the fastener: 4000

The general criteria for the determination of some ones of the above parameters (effective width, distance to the end and fasteners pitch) are explained below:

– Configuration of reference:

The configuration of the test specimens used to obtain the allowable values was the following:

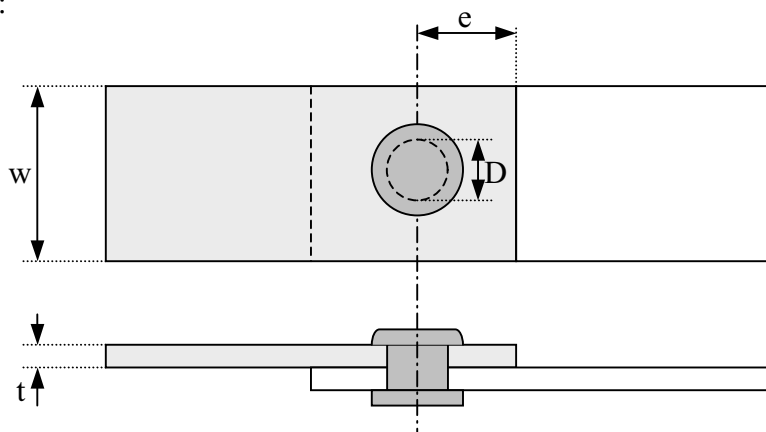


Figure 3: Configuration of reference.

In this case, for the analysis of the upper element, it must be defined:

- = Effective width of the laminate w
- = Distance to the end of the joint in the load direction e
- = Fastener pitch in the load direction N/A

– General configuration:

A more general kind of joint than above is shown in the following figure:

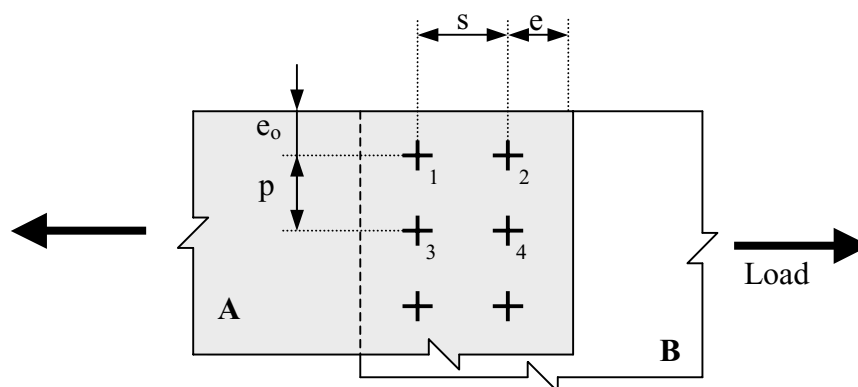


Figure 4: General configuration.

In this case, for the analysis of the upper element (A), the geometrical parameters for each fastener must be defined according to the following table:

Fastener	Effective width	Distance to the end	Fasteners pitch
1	Minimum ($p, 2 \cdot e_o$)	N/A	s
2	Minimum ($p, 2 \cdot e_o$)	e	N/A
3	p	N/A	s
4	p	e	N/A

- Staggered pitch joints transferring axial load:

This kind of joint can be seen in more detail in the following figure:

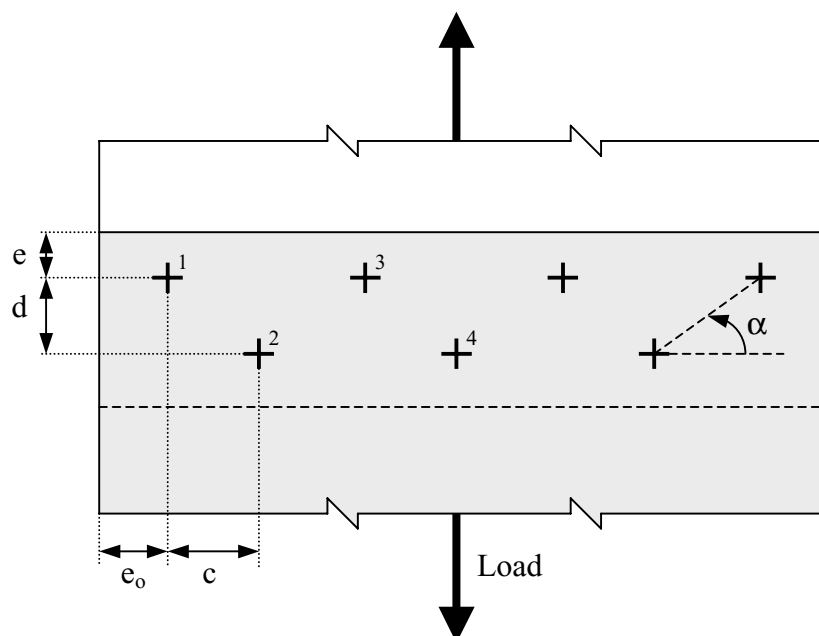


Figure 5: Staggered pitch joint. Axial load.

In this case, for the analysis of the upper element, the geometrical parameters for each fastener must be defined according to the following table:

Fastener	Effective width	Distance to the end	Fasteners pitch
1	Minimum ($c, 2 \cdot e_o$)	e	N/A
2	c	N/A	d
3	c	e	N/A
4	c	N/A	d

NOTE: This criterion is too conservative for staggered patterns with $\alpha > 60^\circ$. In such a case, it can be considered as two parallel rows of fasteners.

– Staggered pitch joints transferring shear:

This type of joint is shown in the following figure:

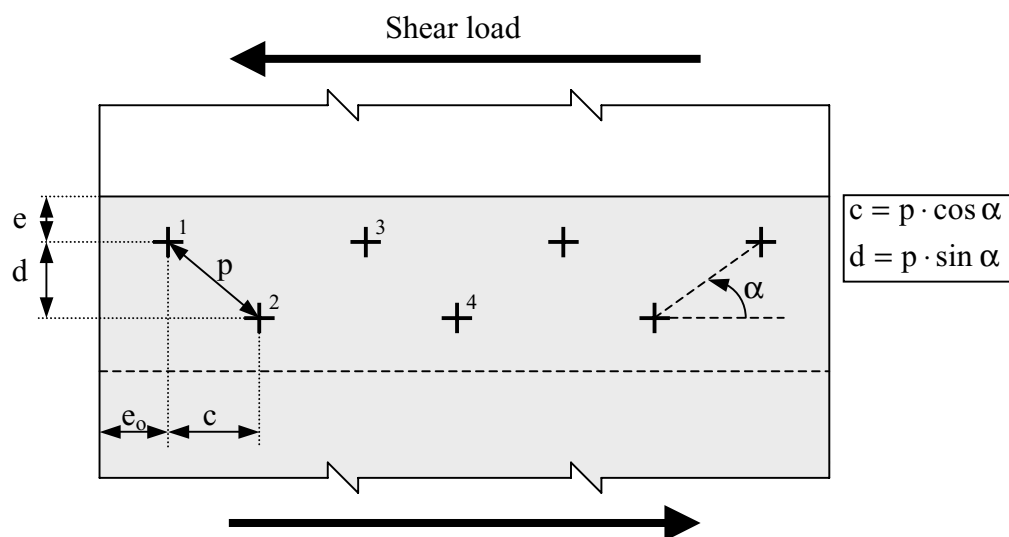


Figure 6: Staggered pitch joint. Shear.

In this case, for the analysis of the upper element, the geometrical parameters for each fastener must be defined according to the following table:

Fastener	Effective width		Distance to the end	Fasteners pitch
	$\alpha < 30^\circ$	$\alpha \geq 30^\circ$		
1	$2 \cdot e$	Minimum ($p, 2 \cdot e$)	e_0	N/A
2	$2 \cdot (d + e)$	p	N/A	c
3	$2 \cdot e$	Minimum ($p, 2 \cdot e$)	N/A	c
4	$2 \cdot (d + e)$	p	N/A	c

NOTE: The results obtained in this case can be conservative or not due to the following reasons:

- = For the same fastener load, the stress concentration due to the shear transfer is, in general, higher than for the axial load transfer (no conservative).
- = The point of higher stress due to bearing is different from the corresponding to by-pass load (conservative).

In all of above cases, when a parameter does **not apply** (N/A) or it is **infinite** (enough big from a practical point of view), a null value (0) must be input to the program.

After this point, the program will perform the analysis of the joint and it will provide the results on the screen.

5 Output results

The URCO program provides the analysis results on the screen, in interactive way. After the input data, the following results are displayed:

- **Input data:**

MATERIAL:

=====

```
1 ... Fabric Group III
2 ... Tape   Group III
3 ... Fabric Group IV
4 ... Tape   Group IV
5 ... Fabric Group V
6 ... Tape   Group V
```

Type of material: 6

Percentage of layers at 0 degrees (%): 25

Percentage of layers at 45 degrees (%): 50

TYPE OF JOINT:

=====

```
1 ... Single shear
2 ... Double shear
```

Type of joint: 1

TYPE OF FASTENER:

=====

```
1 ... Fastener of protuding head
2 ... Countersunk fastener
```

Type of fastener: 2

Fastener diameter: 4

Effective width of the laminate: 16

Distance to the end of the joint in the load direction: 100

Laminate thickness: 4

Fastener pitch in the load direction: 16

By-pass load: + Tension
- Compression

Microstrains due to the by-pass load: 2000

Transferred load through the fastener: 4000

- **Output results:**

They are displayed on the following screen:

```
Allowable values of the joint:
=====

Point A:  5463 microstrains
Point B:   586 N/mm2
Point C:   340 N/mm2

Reserve factors of the joint:
=====

Cross section failure:  Rf = 1.262
Failure due to bearing: Rf = 1.360
```

The output results have the following meaning:

Allowable values:

- **Point A:** It means the tension or compression strength in the cross section at the hole location of the laminate for the by-pass load only.
- **Point B:** It is the tension strength in the cross section at the hole location for the applied load in the fastener.
- **Point C:** It is the bearing strength of the laminate submitted only to the fastener load.

Reserve factors:

- **Cross section failure Rf:** Reserve factor for failure of the cross section at the hole location due to tension or compression.
- **Failure due to bearing Rf:** Reserve factor for bearing failure.

- **Output file:**

It is possible to keep all the above information in a text file. For this purpose, the program asks if the user wants to save the results in a file:

```
Do you want to print the results in an file? ..... (Y/N): Y

Name of the output results file: Example.txt
```

After this screen, the program ends the execution.

- **Errors and warnings:**

When the program detects an error in the input data, it writes the corresponding message in the console and repeats the same question.

If some value in the input data exceeds the range of available test results considered for the development and validation of the program, the corresponding warning is displayed on the screen and the program continues the execution.

The error and warning messages that might appear in the console, are listed below:

Error messages. English output:

```
ERROR: Selection out of bounds.
ERROR: No numerical value.
ERROR: It must be in the range of:
       0 < Percentage(0) < 100.
ERROR: It must be in the range of:
       0 < Percentage(0/90) < 100.
ERROR: It must be in the range of:
       0 < Percentage(45) < 100.
ERROR: It must be:
       Percentage(0) + Percentage(45) < 100.
ERROR: The diameter must be positive.
ERROR: The width must be greater than the diameter.
ERROR: The distance must be greater than the radius.
ERROR: The thickness must be positive.
ERROR: The pitch must be greater than the diameter.
ERROR: The load must be positive.
```

Error messages. Spanish output:

```
ERROR: Selección fuera de límites.
ERROR: Valor no numérico.
ERROR: Debe estar comprendido entre:
       0 < Porcentaje(0) < 100.
ERROR: Debe estar comprendido entre:
       0 < Porcentaje(0/90) < 100.
ERROR: Debe estar comprendido entre:
       0 < Porcentaje(45) < 100.
ERROR: Debe cumplirse que:
       Porcentaje(0) + Porcentaje(45) < 100.
ERROR: El diámetro tiene que ser positivo.
ERROR: La anchura debe ser mayor que el diámetro.
ERROR: La distancia debe ser mayor que el radio.
ERROR: El espesor debe ser positivo.
ERROR: El paso debe ser mayor que el diámetro.
ERROR: La carga debe ser positiva.
```

Warning messages. English output:

WARNING: There are not test results for that value.
A and B will be calculated by extrapolation.

WARNING: There are not test results for that value.
A, B and C will be calculated by extrapolation.

WARNING: There are not test results for this Width/Diameter ratio.
A and B will be calculated by extrapolation.

WARNING: There are not test results for this Distance/Diameter ratio.
B will be calculated by extrapolation.

WARNING: There are not test results for this Thickness/Diameter ratio.
B and C will be calculated by extrapolation.

WARNING: There are not test results for this Thickness/Diameter ratio.
B will be calculated by extrapolation.

WARNING: There are not test results for this Pitch/Diameter ratio.
B will be calculated by extrapolation.

WARNING: The Pitch/Diameter ratio does not satisfy the design principles.

Warning messages. Spanish output:

AVISO: No hay datos de ensayo para ese valor.
A y B serán calculados por extrapolación.

AVISO: No hay datos de ensayo para ese valor.
A, B y C serán calculados por extrapolación.

AVISO: No hay datos de ensayo para esta relación Anchura/Diámetro.
A y B serán calculados por extrapolación.

AVISO: No hay datos de ensayo para esta relación Distancia/Diámetro.
B será calculado por extrapolación.

AVISO: No hay datos de ensayo para esta relación Espesor/Diámetro.
B and C serán calculados por extrapolación.

AVISO: No hay datos de ensayo para esta relación Espesor/Diámetro.
B será calculado por extrapolación.

AVISO: No hay datos de ensayo para esta relación Paso/Diámetro.
B será calculado por extrapolación.

AVISO: La relación Paso/Diámetro no cumple las condiciones de diseño.

6 Windows application

An application has been developed in Visual Basic for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please consult with the system manager.

The application can be executed by "double click" in the file **Urco.exe**, located in the PC's server, or by means of a **link** in the "desktop" or in the "start menu".

The input data must be typed in the application form. The form, after loading the data corresponding to the example (see section 7), has the following appearance:

Calculation of allowable values in composite fastened joints

Length units Millimeters	Fastener diameter D 4.0
Force units Newtons	Effective width W 16.0
Material Tape of Group V	Distance to the end e 100.0
% of layers at 0° 25.0	Laminate thickness t 4.0
% of layers at ±45° 50.0	Fasteners pitch p 16.0
Type of joint Single Shear	By-pass microstrains $\mu\epsilon$ 2000.0
Type of fastener Countersunk	Transferred load P 4000.0

Language:
☐ Español
☒ English

Program execution

Figure 7: Input data form.

The program can be executed clicking the button "Program execution". Then the application solves the problem and shows the results in the following form:

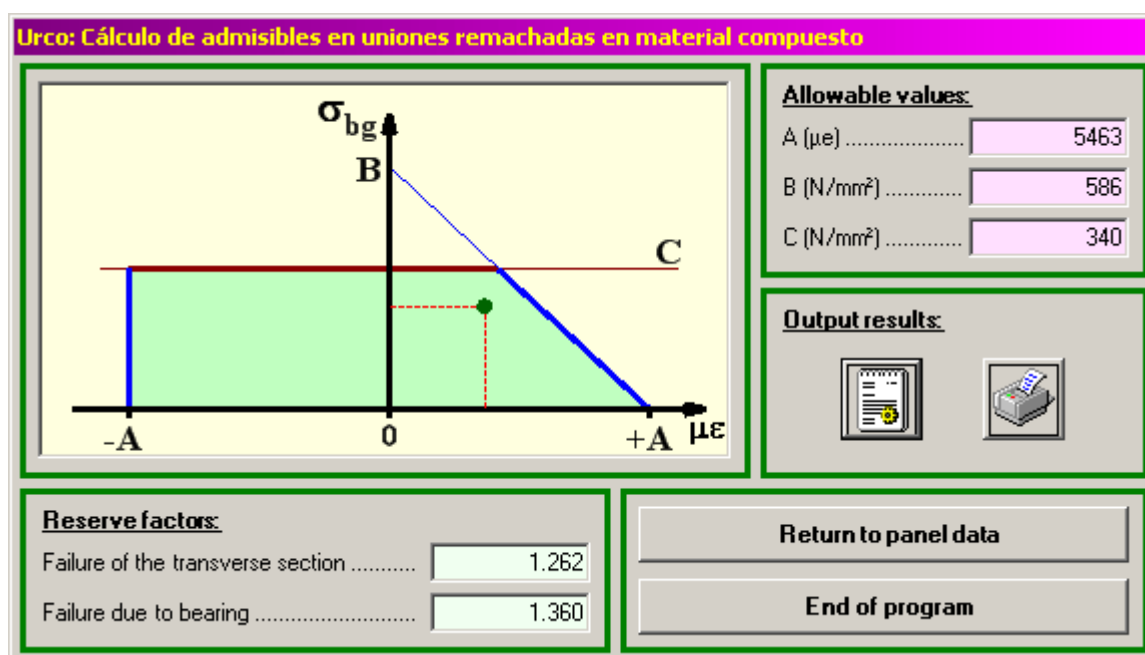


Figure 8: Output data form.

The buttons located under the label “Output results:” are to write the results in a file (left button) or in a printer (right button). The file obtained pressing the corresponding button has the same format as the one provided by the official program.

7 Example

The case to be solved consists in an axial joint between two plates, fastened each other, as shown in the following figure:

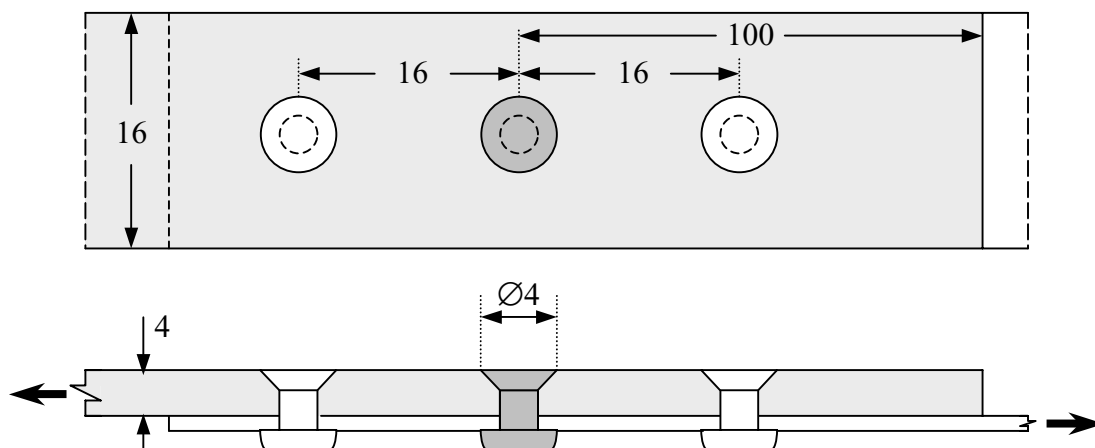


Figure 9: Example geometry.

The element to be studied, the upper one, is supposed to be made in 8552 / AS4 tape (group V material), with a typical orthotropic lay-up (25/50/25).

The hole under study will be the central one. The load transferred through the fastener as well as the by-pass stress/strain at the hole must be determined by means of a load transfer analysis (i.e. running the ARES program).

In this example, the following loading will be assumed:

By-pass strain, ϵ2000 $\mu\epsilon$

Transferred load, P4000 N

With all the above data, now it is possible to run the program. The screen data flow can be the following, starting by the language selection:

```
Idioma / Language:
=====
1 ... ESPAÑOL
2 ... ENGLISH
Idioma / Language: 2
```

The next step is to define the material and the lay-up:

MATERIAL:

=====

- 1 ... Fabric Group III
- 2 ... Tape Group III
- 3 ... Fabric Group IV
- 4 ... Tape Group IV
- 5 ... Fabric Group V
- 6 ... Tape Group V

Type of material: **6**

Percentage of layers at 0 degrees (%): **25**

Percentage of layers at 45 degrees (%): **50**

Looking at the geometry of the problem it can be seen that it is a single shear joint with countersunk fasteners. Therefore, the following data (all units are “mm” and “N”) must be input in the console, as shown below:

TYPE OF JOINT:

=====

- 1 ... Single shear
- 2 ... Double shear

Type of joint: **1**

TYPE OF FASTENER:

=====

- 1 ... Fastener of protuding head
- 2 ... Countersunk fastener

Type of fastener: **2**

Fastener diameter: **4**

Effective width of the laminate: **16**

Distance to the end of the joint in the load direction: **100**

Laminate thickness: **4**

Fastener pitch in the load direction: **16**

By-pass load: + Tension
- Compression

Microstrains due to the by-pass load: **2000**

Transferred load through the fastener: **4000**

Name of the output results file: **Example.txt**

Material	Tape Group V
% of layers of fibres at 0° ...	25.00
% of layers of fibres at 45° ...	50.00
% of layers of fibres at 90° ...	25.00

Output results file: (cont.)

Joint data:

=====

Type of joint: Single Shear

Type of fastener: Countersunk

Geometry of the joint:

=====

Fastener diameter: D = 4.00 mm

Effective width: W = 16.00 mm

Distance to the end: e = 100.00 mm

Laminate thickness: t = 4.00 mm

Fasteners pitch: p = 16.00 mm

Load case:

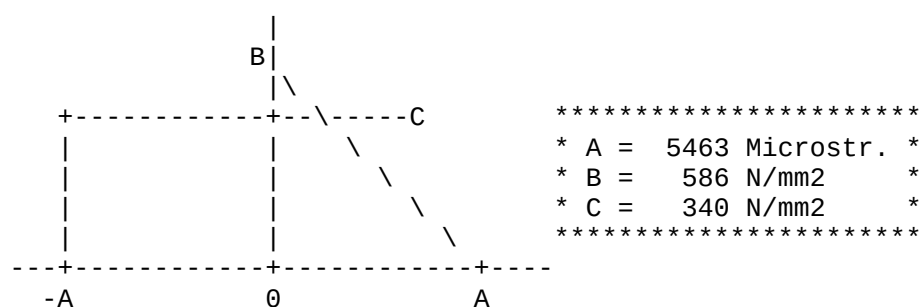
=====

By-pass microstrains: s = 2000

Transferred load: P = 4000 N

Allowable values:

=====



Reserve factors:

=====

Cross section failure: Rf = 1.262

Failure due to bearing: Rf = 1.360

Program Urco. Version 4.1. (c) EADS CASA 2006
